Free pythagorean theorem worksheet worksheet 12

Maths | 9th grade | Mixed Operations

1. If 3x - 4y = 5 and x + 2y = 3, find the value of 5x + 6y.

A) 14
B) 16
C) 12
D) 10

2. Evaluate: (3^3 - 2^3) / (5 - 2)

A) 7
B) 9
C) 5
D) 11

3. If f(x) = 2x^2 + 3x - 1, find f(-1).

A) 0
B) -6
C) 1
D) -2

4. Simplify: (2x^2y^3)^2 * (xy^2)^3.

A) 4x^7y^12
B) 4x^8y^12
C) 8x^8y^12
D) 4x^8y^10

5. Find the sum of the roots of the quadratic equation: x^2 - 7x + 10 = 0.

A) 7
B) 5
C) -7
D) -10

6. Solve for x: 5^(2x-1) = 125.

A) 2
B) 3
C) 1
D) 4

7. If log_2(x) + log_2(8) = 5, what is x?

A) 32
B) 16
C) 8
D) 4

8. The sequence 4, 8, 16, ... is geometric. Find the 6th term.

A) 128
B) 256

C) 512
D) 64

9. Calculate the derivative of f(x) = x^3 - 4x^2 + 6x - 5 at x = 2.

A) 5
B) 3
C) 7
D) 9

10. If the area of a triangle is 30 square units and its base is 10 units, find its height.

A) 6
B) 7
C) 5
D) 8

Answer Key

1. B
Solve the system of equations to find x and y, then substitute into 5x + 6y.

2. A
Calculate the power values, subtract, then divide by 3.

3. B
Substitute x = -1 into the function and simplify.

4. B
Apply power rules and multiply powers of similar bases.

5. A
Use the sum of roots formula: -b/a for ax^2 + bx + c = 0.

6. A
Express 125 as a power of 5 and equate the exponents.

7. A
Use the log property to combine terms and solve for x.

8. C
Use the formula for the nth term of a geometric sequence: ar^(n-1).

9. D
Differentiate the function and substitute x = 2.

10. C
Use the area formula for a triangle: 0.5 * base * height = area.